

SPECIAL NOTE: You must show all of your work to receive full credit. Do all calculations in the exam booklet provided. To receive full credit, write your final answer in the spaces provide on this sheet. You must turn this sheet in with your exam booklet.

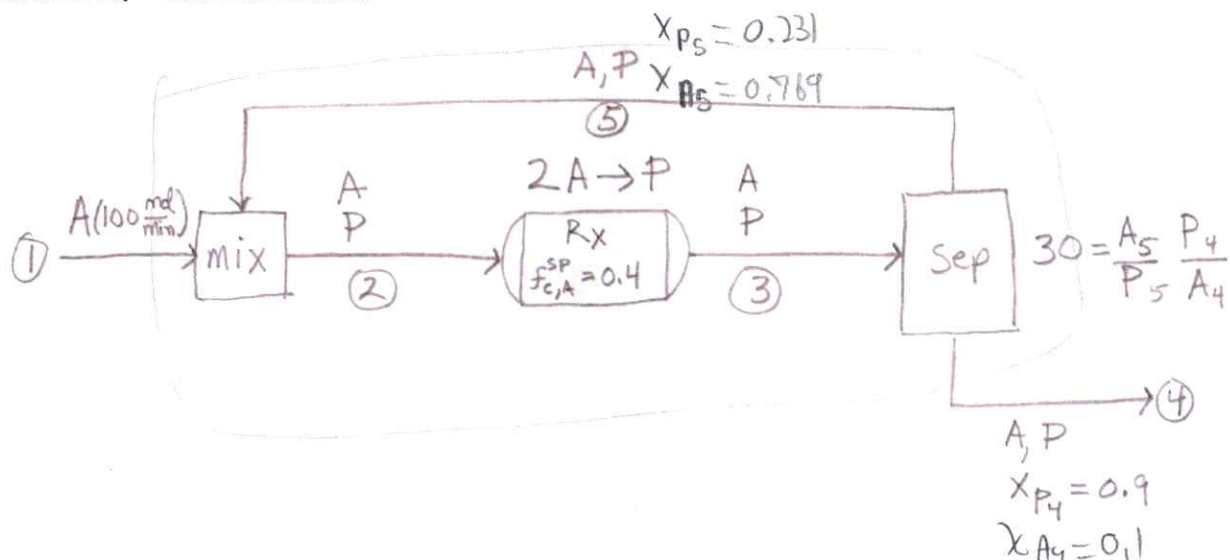
- (1) (50 pts.) Consider the block flow diagram below. Notice that the streams are labeled 1 through 5. Species molar flow rates are given by the species symbol with the stream number as a subscript. The fresh feed contains only species A. This stream is mixed with a recycle stream containing A and P where P is the reaction product. The mixture enters a reactor in which one reaction takes place:



The reactor converts 40% of A entering the reactor. The outlet stream from the reactor goes to a separator. Both streams leaving the separator are in equilibrium containing some A and P such that:

$$\alpha_{AB} = \frac{A_5 P_4}{P_5 A_4}$$

where the separation factor $\alpha_{AB} = 30$. Stream 5 is the stream enriched in A and is recycled back to the reactor. Stream 4 is enriched in the product P such that it is 90 mol% P. The fresh feed has species flow rates of $A_1 = 100$ mol A/min.



- (a) Find all of the species molar flow rates not given in the problem statement. Fill in the table below with the appropriate species flow rates in mole/min to receive full credit. (45 pts.)

Species↓/Stream→	1	2	3	4	5
A	100			$0.1 \dot{n}_4$	$0.769 \dot{n}_5$
P	----			$0.9 \dot{n}_4$	$0.231 \dot{n}_5$

- (b) Calculate the overall fractional conversion based on species A (5 pts.)

$$f_{c,A}^{ov} = \frac{A_1}{A_1}$$

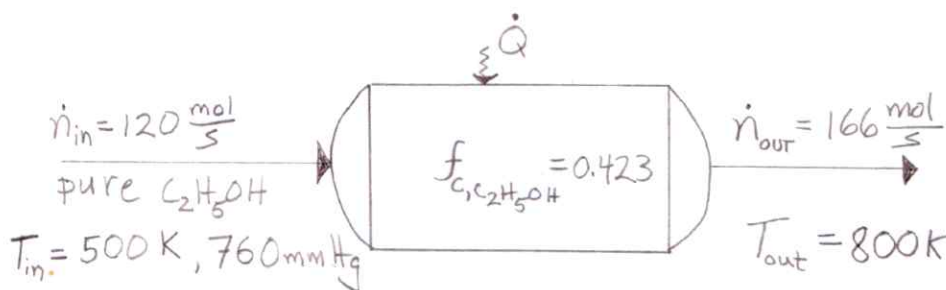
(2) (50 pts.) Acetaldehyde (CH_3CHO) is produced by dehydrogenation of ethanol ($\text{C}_2\text{H}_5\text{OH}$).



An undesired side reaction produces ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$)



This process is carried out in a single reactor operating at steady state. The reactor has a feed rate of 120 mol/s of pure ethanol at 500K and 760 mm Hg. The reactor total product flow rate is 166 mol/s at 800K. The reactor converts 42.3% of the ethanol.



Data: The heat capacity in J/mol°C is given by $C_p = A + Bt$ where $t = T(\text{K})/1000$. The table below gives the constants A and B .

Species	A	B
$\text{C}_2\text{H}_5\text{OH}$	32.3	120
CH_3CHO	30.7	88.2
$\text{CH}_3\text{COOC}_2\text{H}_5$	60.9	191
H_2	28.4	1.54

- (a) Calculate the all the species molar flow rates for the product stream in mol/s and write them into the Input/Output Table below. Also, calculate the two extents of reaction, ξ_1 and ξ_2 . (20 pts.)

$$\xi_1 = 41.24 \text{ mol/s} \quad \xi_2 = 4.76 \text{ mol/s}$$

- (b) Finish filling enthalpy values into the Input/Output Table below. Use a reference state of $T_{\text{ref}} = 25^\circ\text{C}$. (10 pts.)

	Species	$\dot{n}_{in}$ (mol/s)	$\bar{H}_{in}$ (kJ/mol)	$\dot{n}_{out}$ (mol/s)	$\bar{H}_{out}$ (kJ/mol)
E	$\text{C}_2\text{H}_5\text{OH}$	120	$H_a = 0.016496$	69.24	$H_1 = 0.049286$
A	CH_3CHO	----	----	41.24	$H_2 = 0.039719$
EA	$\text{CH}_3\text{COOC}_2\text{H}_5$	----	----	4.76	$H_3 = 0.083211$
H	H_2	----	----	50.76	$H_4 = 0.014681$

- (c) Calculate the heat rate in kJ/s for the reactor. Indicate clearly whether it is heat in or out of the reactor. (20 pts.)

$$\dot{Q} = 2960.402 \text{ kJ/s}$$

into reactor

I.D. #

Name

Subject

Course

Section

Instructor

Date

Jacob Golota

1	31
2	48
1	79

Receiving or giving aid in a final examination is a cause for dismissal from the University.

(Begin Writing on this page)

①

Mix

$$A: A_2 = 100 + A_5$$

$$P: P_3 = P_5$$

$$A_1 = 100$$

$$A_3 = (1 - 0.4)A_2$$

$$A_3 = 0.6A_2$$

All

$$A: A_4 = 100 - 2\$$$

$$P: P_4 = 0 + \$$$

$$P_4 = \$$$

Router

$$A: A_3 = A_2 - 2\$$$

$$P: P_3 = P_2 + \$$$

Sep

$$A: A_5 + A_4 = A_3$$

$$P: P_5 + P_4 = P_3$$

$$0.6A_2 = P_3 - 2\$$$

$$2\$ = 0.4A_2$$

$$\$ = 0.2A_2$$

② $E: E_2 = 120 \frac{\text{mol}}{\text{s}} - S_1 - 2S_2 \quad ; \quad E_2 = 120 - S_1 - 2S_2$

$H: H_2 = 0 + S_1 + 2S_2 \quad ; \quad H_2 = S_1 + 2S_2$

$EA: EA_2 = 0 + S_2 \quad ; \quad EA_2 = S_2$

$A: A_2 = 0 + S_1 \quad ; \quad A_2 = S_1$

$E_2 = (1 - 0.4123)E_1$

$E_2 = 0.5877 \times E_1 = 69.124 \frac{\text{mol}}{\text{s}}$

$\begin{cases} 69.124 = 120 - S_1 - 2S_2 \\ S_1 + 2S_2 = 50.876 = H_2 \end{cases}$

$\dot{n}_{out} = 166 \frac{\text{mol}}{\text{s}} = E_2 + H_2 + EA_2 + A_2$

$46 = EA_2 + A_2$

$46 = S_2 + S_1$

$50.876 = 2S_2 + S_1$

$4.176 = S_2$

$\begin{cases} 46 = 4.176 + S_1 \\ 41.24 = S_1 \end{cases}$

20/20

$E_1 H_1 = \int_{0.248}^{0.8} 32.3 + 120t = 49.286 \frac{\text{J}}{\text{mol}}$

$A H_2 = \int_{0.248}^{0.8} 30.7 + 88.2t = 39.719 \frac{\text{J}}{\text{mol}}$

$EA H_3 = \int_{0.248}^{0.8} 60.9 + 191t = 83.211 \frac{\text{J}}{\text{mol}}$

$H H_4 = \int_{0.248}^{0.8} 28.4 + 1.54t = 14.681 \frac{\text{J}}{\text{mol}}$

$E_a H_a = \int_{0.248}^{0.5} 32.3 + 120t = 16.196 \frac{\text{J}}{\text{mol}}$

$\begin{cases} T_1 = 25^\circ\text{C} + 273 = 298 \\ t_1 = 0.248 \end{cases}$

$\begin{cases} T_2 = 800^\circ\text{C} \\ t_2 = 0.8 \end{cases}$

$\begin{cases} T_3 = 500^\circ\text{C} \\ t_3 = 0.5 \end{cases}$

$\rightarrow 0.049286 \frac{\text{kJ}}{\text{mol}}$

$\rightarrow 0.039719 \frac{\text{kJ}}{\text{mol}}$

$\rightarrow 0.083211 \frac{\text{kJ}}{\text{mol}}$

$\rightarrow 0.014681$

$\rightarrow 0.016196$

x1000 units

8/10

$$c) Q = \sum \Delta \hat{H}_f^\circ \nu_i + \sum_{out} \Delta \hat{H}_i \dot{n}_i - \sum_{in} \Delta \hat{H}_i \dot{n}_i$$

$$Q = \Delta \hat{H}_{f1}^\circ \dot{g}_1 + \Delta \hat{H}_{f2}^\circ \dot{g}_2 + [(69.24)(0.049286) + (41.24)(0.03996) + (4.76) \times (0.083211) + (50.76)(0.04127)] - [100](0.06146)$$

$$Q = 41.24 \Delta \hat{H}_{f1}^\circ + 4.76 \Delta \hat{H}_{f2}^\circ + 4.248 \text{ kJ/s}$$

$$\begin{aligned} \Delta \hat{H}_{f1}^\circ &= (-1)(H_{F,E}) + (1)(H_{F,A}) + (1)(H_{F,H}) \\ &= (-1)(-234.95) + (1)(-166.2) + (1)(0) \\ &= 68.75 \text{ kJ} \end{aligned}$$

$$\begin{aligned} \Delta \hat{H}_{f2}^\circ &= (-2)(H_{F,E}) + (1)(H_{F,EA}) + (2)(H_{F,H}) \\ &= (-2)(-234.95) + (1)(-444.5) \\ &= 25.4 \end{aligned}$$

$$\begin{aligned} Q &= 2835.25 \frac{\text{kJ}}{\text{s}} + 120.904 \frac{\text{kJ}}{\text{s}} + 4.248 \frac{\text{kJ}}{\text{s}} \\ &= 2960.402 \frac{\text{kJ}}{\text{s}} \end{aligned}$$

20/20

44/50

① $A_1 = 100 \frac{\text{mol}}{\text{min}}$, $f_{C,A} = 0.4$, $\Delta A_B = 30$

Mix	Reactor	Sep	Over All
$A: A_2 = 100 + A_5$	$A: A_3 = A_2 - 2S$	$A: A_4 + A_5 = A_3$	$A: A_4 = 100 - 2S$
$P: P_2 = P_5$	$P: P_3 = P_2 + S$	$P: P_4 + P_5 = P_3$	$P: P_4 = 0 + S$

$f_{C,A} = 0.4 \rightarrow A_3 = (1 - 0.4)A_2$

$A_3 = 0.6A_2$

DOF

Mix: $(5 + 0) - (1 + 0 + 0 + 2) = 5 - 3 = 2$

Reactor: $(4 + 1) - (0 + 0 + 1 + 2) = 5 - 3 = 2$

Sep: $(6 + 0) - (0 + 1 + 1 + 2) = 6 - 4 = 2$

Over All: $(9 + 1) - (1 + 1 + 2 + 2) = 10 - 6 = 4$

$30 = \frac{A_5}{P_5} \frac{P_4}{A_4}$

$P_4 = x_{P4} n_4 = 0.9 n_4$

$A_4 = 0.1 n_4$

$30 = \frac{A_5}{P_5} \left(\frac{0.9 n_4}{0.1 n_4} \right)$

$3.3 = \frac{A_5}{P_5}$

$A_5 = 3.33 P_5$

$0.6 A_5 = P_5$

$A_2 = 100 + 0.98 n_5$

$x_{P4} = 0.9$

$x_{P5} = 0.2308$

$x_{A4} = 0.1$

$x_{A5} = 0.769$

$x_{A5} = \frac{A_5}{A_5 + P_5} = \frac{A_5}{A_5 + 0.6 A_5} = \frac{A_5}{1.6 A_5} = 0.625$

$x_{P5} = 0.2308$

yes!

$$A_3 = A_4 + A_5$$

$$A_3 = 0.1n_4 + 0.769n_5$$

$$P_3 = 0.9n_4 + 0.231n_5$$

$$A_3 \times 0.16 A_2$$

$$0.1A_2 \times \frac{1}{2} = P_3$$

$$0.2A_2 = P_3 = P_5 + \$$$

$$0.2A_2 = P_5 + \$$$

$$0.2(100 + A_2) = P_5 + \$$$

$$20 + 0.2A_2 = P_5 + \$$$

$$20 + 0.2(0.769n_5) = (0.231n_4) + \$$$

$$20 + 0.1538n_5 = 0.231n_5 + \$$$

$$20 = 0.0769n_5 + \$$$

bro im lost on question one, Q2 was good

8/15

21/30

2/5

31/50